

Do Voters Respond to the Economy or to News Reporting on the Economy? A Mediation Analysis

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That the economy influences support for the government is a cornerstone of electoral research, but how this comes about is less understood. Largely due to data limitations, researchers have had difficulty estimating how much of the economy's effect on the vote is direct and how much is driven by media coverage. New data and automated text analysis make possible the first media-oriented mediation analysis of the economic vote. Using 3 million news articles from 15 developed countries in six languages, we demonstrate that about 30% of the effect of growth is mediated through news sentiment. In contrast, the effect of unemployment on elections is direct. Deviations of economic reporting from the actual economy correlate with election outcomes, though not as much as is sometimes presumed, as we detail in two election examples in which media reporting likely did and did not determine the outcome.

Unfair media coverage is a common complaint among politicians. After his party's embarrassing defeat in the 2019 UK parliamentary elections, Labour leader Jeremy Corbyn reacted to coverage of himself in the press—"This Corbyn guy, God he's evil. I wouldn't want to live in the same street as him" (Osborne 2020). Indeed, recent scholarship has emphasized the media's role in elections. DellaVigna and Kaplan (2007) and Enikolopov, Petrova, and Zhuravskaya (2011) examined the general tendency of slanted news to persuade voters in elections. Ladd and Lenz (2009) and Chiang and Knight (2011) studied the ability of the media to persuade voters through explicit endorsements.

While Corbyn's complaint posits and these studies demonstrate that valenced reporting may persuade voters, an alternative pathway for media influence is through coverage that touches on objective events and indicators. Examples include a country's loss in a military conflict, an increase in the crime rate, and a decrease in the unemployment rate. In each case, media coverage will be grounded in an objective truth (e.g., the unemployment rate really did decrease), but the coverage will generally go beyond the objective facts (Stevenson and Duch 2013) due to the desire of the media to influence

perceptions, biases in how reporters process information, or the constraints of the medium. The need to write complete newspaper articles or to build complete television segments provides a potential way in which the media may affect election outcomes—by coloring objective facts or indicators. The news media may alter voters' perceptions of the event or indicator, which may ultimately alter the same voters' views of the candidates for office.

The importance of the economy to election outcomes has regularly drawn attention to whether media coverage accurately portrays the economy. Italian prime minister Silvio Berlusconi frequently complained of a media that exaggerated economic difficulties (Reuters 2009). Former President George H. W. Bush similarly attributed his reelection loss to inaccurate media coverage of the economy, with scholars such as Lipset (1993) and Hetherington (1996) arguing that his perception was correct. More generally, scholars such as Sanders and Gavin (2004) and Boomgaarden et al. (2011) have established the role of the media in economic perceptions, with the former arguing that the actual economy has little effect after accounting for media coverage. De Boef and Kellstedt (2004) and Hollanders and Vliegenthart (2011) assert that the

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Replication files are available in the *JOP* Dataverse (<https://dataverse.harvard.edu/dataverse/jop>). The empirical analysis has been successfully replicated by the *JOP* replication analyst. An online appendix with supplementary material is available at <https://doi.org/10.1086/734534>.

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media influences consumer confidence, whereas Goidel et al. (2010) find that the media influences individuals' perceptions of their own financial situation.

If voters indeed hold elected officials accountable for media misrepresentations to any large degree, not only would it yield officials elected on false pretenses but it would erode the quality of democratic governance in general. Scholars have grappled with the issue of unjust electoral rewards and punishments for economic outcomes that originate from global economic shocks or other sources beyond the influence of politicians (Campello and Zucco 2020; Duch and Stevenson 2008; Hart and Matthews 2022; Hellwig and Samuels 2007). False accountability for systematic inaccuracies in economic news could pose an even more fundamental threat to democratic accountability. To what degree, then, can the media harm a candidate politically through inaccurate or misleading reporting of objective facts? This depends on how voters learn about those facts. If voters learn about the economy primarily through personal experience, inaccurate media coverage may ultimately not harm these politicians. Conversely, if media coverage is the main way that voters learn about the economy, they may be persuaded by inaccurate or unfair media coverage. Missing from existing work is an estimate of the size of the role of media sentiment, if any, in the economic vote. This lack of attention can be understood by considering the difficulty of collecting measures of media content.

In order to measure the relative effect on the vote of (1) the economy and (2) media reporting about the economy, one has to collect long economic and news media time series. Because economic news time series depend on digital news archives that often do not go back far in time and each country only has one national economy, precise estimation of economic and media effects requires collecting economic and economic news data from multiple countries. Previous economic news data collection has typically considered a single English-speaking country,¹ limiting the ability to perform mediation analysis.

To study the media's role in the economic vote, we employ a dataset of 3 million newspaper articles related to the economy from 30 newspapers in six languages in 15 developed democracies. Measuring media reporting is challenging because media coverage is fragmented and its nature has varied over time. Voters can learn about the economy through newspapers, television, radio, websites, and social media. An additional challenge is that our analysis necessitates a fairly long time series of media coverage. Newspapers offer us the best

chance of capturing the media environment over an extended period of time for two reasons. First, newspapers offer the longest period of coverage from the various media sources available. The availability of television transcripts starts later, with social media being relevant only in the most recent time periods. Second, newspapers are the best single candidate for measuring the overall media environment. Roberts and McCombs (1994) demonstrate that newspapers lead other news sources. Druckman (2005) demonstrates that the content of newspaper coverage differs little from the content of television coverage. Maier (2010) similarly finds commonality between the coverage of television and print media as well as commonality in the coverage in web news. The period of time we study (1977–2014) sees television and print news as the top news sources, and even in the later years we study, social media remains a minor source of news (Shearer 2018).

From our sample of newspapers, we generate a measure of the media's tone in covering three aspects of the economy—growth, unemployment, and inflation. We then estimate models that include measures of both economic performance and the media's coverage of economic performance as explanatory variables and apply mediation analysis to quantify the indirect effect of the economy (the part of the economic vote that is due to the correlation between the actual economy and media coverage of the economy) and the direct effect of the economy (the part of the economic vote that is not colored by media coverage). Because media reports that perfectly match the performance of the economy are captured by the direct effect, what the indirect effect reports is the effect of media reporting that deviates from the actual performance of the economy.

Our results indicate that the effect of economic growth is partially mediated by news coverage—approximately 30% of the total effect of growth in elections is indirect, stemming from media sentiment. In contrast, we find that the effect of unemployment is entirely direct—we find no evidence that unemployment sentiment in the news has a mediating effect and the ratio of the direct effect to the total effect is close to one and precisely estimated. Inflation has an indeterminate effect on elections—we find some marginal evidence for a total effect for inflation, but our main specifications suggest that the ratio of the indirect effect to the total effect is near zero. These results for inflation are, however, less robust to alternative specifications than our results for growth and unemployment. Ultimately, our results suggest that media sentiment is related in a detectable, but limited way, to the fortunes of incumbent governments—substantial enough to predict a change in election outcomes in rare cases when deviations between reporting and economic performance are large and vote margins very small, but unlikely to entail a change in outcomes for most elections. We illustrate this with case studies of two elections

1. One exception to this is Stevenson and Duch (2013), who collected data from the front pages of leading newspapers in six countries between 1980 and 2000.

with large gaps between economic news and economic performance—the Australian general election of 2010 and the US presidential election of 1992.

ECONOMIC VOTE MECHANISMS

Multiple mechanisms could underpin the economic vote but the absence of sufficient data on economic news has long complicated the testing of one of the most important—the mediating role of economic sentiment in the media. Voters may (1) learn about the economy through personal experience, for example by losing a job, receiving a large raise, hearing about a friend losing their job, or seeing a local business close. Alternatively, voters may (2) respond to their perceptions of the national economy rather than their personal experience. Media sentiment on the economy enables this second mechanism by contextualizing economic statistics when reporting on them in ways that may deviate from actual performance. Finally, voters may (3) possess actual knowledge of the national economy. Although few voters consume economic statistics directly from official sources, others might be able to decontextualize news coverage and arrive at a sense of the national economy, uncolored by the ideological and behavioral biases of the media.²

Direct and indirect effects

The key goal of this article is to separate out the component of the economic vote that is driven by contextualized media coverage. To capture this component, mediation analysis estimates a model that includes media sentiment, controlling for a measure of the national economy. The decomposition separates the effect of the national economy into an indirect effect and a direct effect. The indirect effect captures the part of the economic vote that is colored by media sentiment, whereas the direct effect captures the part that is not colored by media sentiment, including, of course, uncolored or decontextualized economic news.

The three plausible mechanisms behind the economic vote—personal experience with the economy, economic perceptions influenced by media sentiment, and knowledge of the national economy—map onto the direct and indirect effects. Consider first how personal experience maps onto the direct effect. The personal experience of a voter could include receiving a raise or losing a job, but it could also include hearing of an acquaintance's lost job, seeing a business close,

or seeing higher prices in a grocery store. Even broadly construed, a voter's personal experience will measure the state of the national economy with error because the voter must sample from available information. However, knowledge of the personal experiences of friends and acquaintances allows voters to sample more information. Ansolabehere, Meredith, and Snowberg (2014) support this notion by demonstrating that individuals from groups that experience greater levels of unemployment report a higher national unemployment rate.

The effect of media sentiment, in contrast, maps onto the indirect effect. Economic news may be colored for a variety of reasons—media outlets may strive to influence voters by reporting on the economy in a biased way, and reporters may be subject to various psychological biases. The media may color raw economic numbers with particular verbs, adjectives, or adverbs, as in “the economy grew a disappointing 2.8% this quarter.” The constraints of the medium may privilege the coverage of negative information as it may attract more readers. The media may be working off of initial economic estimates by government offices that are less accurate than revisions (Croushore 2011) and receive a disproportionate amount of attention in the press (Abo-Zaid 2014). Colored media sentiment can often arise through placing news reports in a specific context. News outlets may selectively report economic statistics by choosing a particular time period or a particular aggregate, or they may color actual economic performance by comparing it to other countries (Hansen, Olsen, and Bech 2015; Hart and Matthews 2022; Kayser and Peress 2012; Park 2019) or to previous time periods (Aytac 2018; Park 2019).

The third mechanism, knowledge of the national economy, maps onto the direct effect of the economy but only to the extent that it is accurate. Voters may learn about the national economy from economic statistics from official sources, through their social networks, or through media coverage of the economy. For media coverage of the economy to form part of the direct effect however, voters must be able to adjust for any deviation in sentiment from the actual performance of the economy. If voters react to colored depictions (as captured by our sentiment measures), this will form the indirect rather than the direct effect. Some scholars argue that voters are indeed able to compensate for ideological biases in newspaper reporting (Chiang and Knight 2011), perhaps by drawing on their personal experience (Mutz 1994).

Interpreting the effects

Whether and to what degree citizens judge their elected officials based on objective economic outcomes as opposed to media misrepresentations matters substantively because the ability of voters to hold their elected representatives accountable for

2. Discussion with other individuals may provide another way that voters learn about the economy, but this can best be thought of as an intermediate pathway—most of the information transmitted through social networks can be characterized as the experiences of others, media sentiment, or knowledge of the national economy.

their performance is a cornerstone of democratic governance (Powell 2000; Przeworski, Stokes, and Manin 1999). One of the most important issues—one that perennially ranks among the most important for voters—is the economy (Heffington, Park, and Williams 2017). If (1) economic news influences vote choice and if (2) media accounts of economic performance deviate from actual performance, then voters may punish or reward their elected officials for economic perceptions that do not match economic performance, tipping democratic accountability in general into question.³ The literature, indeed, includes some evidence for each of these conditions.

Several studies find a strong role for the media in vote choice or other variables relevant to voting such as executive approval and economic perceptions. Erikson, MacKuen, and Stimson (2002) and MacKuen, Erikson, and Stimson (1992) find that (future) consumer sentiment, which is strongly influenced by the media, wholly mediates the effect of economic conditions on presidential approval. De Boef and Kellstedt (2004) and Hollanders and Vliegenthart (2011) argue that media coverage exerts an effect on consumer sentiment, controlling for economic conditions. Negative economic news, even when unbiased, is associated with a decline in expectations (Goidel et al. 2010), expected changes in economic conditions (Boomgaarden et al. 2011; Goidel et al. 2010), and economic perceptions (Garz 2018). Mutz (1994) argues that voters use the media to assess the economy but adjust for bias by using personal experience. Garz and Martin (2021) find that milestone numbers for unemployment, which get reported more in the media, have a disproportionately large effect on the vote shares of incumbent US governors. Duch and Stevenson (2008) argue that “the media plays an integral role in shaping economic evaluations,” which enables voters to compare variances in economic performance across countries. These studies, while allowing for media reporting that deviates from economic performance, primarily focus on media reporting on the economy rather than media bias.

For economic news to undermine accountability, however, not only must it influence voting but it must deviate from the actual performance of the economy. Bias can occur in voters’ comprehension of economic information but also in the economic information itself. Healy and Lenz (2014), for example, demonstrate that voters are subject to recency bias, evaluating the incumbent primarily on economic performance in the year before the election. Wlezien, Franklin, and Twigg (1997) and Evans and Anderson (2006) document

how partisan preferences influence perceptions of the economy. Actual bias in the economic news itself, however, is also possible. Soroka (2006, 2012) assesses negativity bias in economic news, showing that poor economic aggregates receive greater attention. Partisan news extending beyond the economy can influence the strength of partisan attitudes (Hopkins and Ladd 2014) and voting, both by the electorate (Della-Vigna and Kaplan 2007) and their legislative representatives (Arceneaux et al. 2015). Larcinese, Puglisi, and Snyder (2011) demonstrate that US newspapers aligned with the Democratic Party publish more articles on high unemployment under Republican presidents than under Democratic presidents. Conversely, Boydston, Highton, and Linn (2018) argue that most of the variance in media sentiment is explained by the actual economy, whereas Kayser and Peress (2021) argue that media sentiment on the economy is subject to little partisan bias.

The absence of an empirical estimate of the indirect effect of media sentiment in the economic vote leaves open the possibility that inaccurate or biased economic news could sway election outcomes and undermine the ability of voters to control elected officials. Our finding of only a limited role for media sentiment in the economic vote should temper such concerns.

Measuring the mechanisms

Collecting newspaper articles for multiple countries over time allows us to construct a measure of media sentiment on the economy that, together with macroeconomic time series and voting data, enable the estimation of direct and indirect effects in a mediation model. We preview our approach here in a generalized and simplified form to orient readers to our estimation strategy before developing the full later in the manuscript.⁴

Consider two explanatory variables—media sentiment on the economy (S_n) and a measure of the actual economy based on economic statistics (E_n)—and an outcome variable—support for the incumbent prime minister’s party, measured using either election results or vote intentions (V_n). Mediation analysis allows us to distinguish direct and indirect effects through the combination of two equations and the joint interpretation of coefficients. The first model in this approach regresses support for the incumbent government on media sentiment and the actual economy:

$$V_n = \beta_0 + \beta_1 E_n + \beta_2 S_n + \epsilon_n. \quad (1)$$

3. Scholars have found minimal electoral accountability for many issues (Hanretty, Mellon, and English 2021), but a finding for the economy, which is both a valence issue and ranked among the most important to voters, would raise questions about democratic accountability in general.

4. In particular, here, we consider a single measure of the actual economy and a single measure of media sentiment. Later in the article, we discuss a model with growth, unemployment, and inflation and corresponding sentiment measures for each.

The second model regresses media sentiment on the actual economy:

$$S_n = \gamma_0 + \gamma_1 E_n + \eta_n. \quad (2)$$

Mediation analysis proceeds by plugging equation 2 into equation 1 to obtain

$$V_n = \beta_0 + \gamma_0 \beta_2 + (\beta_1 + \gamma_1 \beta_2) E_n + (\epsilon_n + \beta_2 \eta_n). \quad (3)$$

Here, $\beta_1 + \gamma_1 \beta_2$ represents the total effect of the economy on support for the incumbent, which is analogous to the effect of the economy one would find in classical studies of the economic vote. The effect is decomposed into two parts. The direct effect— β_1 —is made up of voters' direct knowledge of the economy and voters' knowledge of the economy that has been gleaned from the media but filtered of media spin. The indirect effect— $\gamma_1 \beta_2$ —captures that part of the economic vote that is driven by media coverage. γ_1 captures the effect of the actual economy on media coverage. β_2 by itself captures the effect of media coverage on the vote. Finally, $(1 - \gamma_1) \beta_2$ is the effect of the inaccurate component of media sentiment. Media coverage does not perfectly reflect the economy, and this captures the distorting effect of media coverage on the vote that has the potential to undermine democratic accountability.

DATA

In this section, we describe the data that we use to perform the mediation analysis. We combine economic sentiment data derived from 30 mainstream newspapers in 15 countries with objective economic indicators and vote share data from elections.

Newspaper articles

Motivated by studies of the economic vote, we sought to collect newspaper articles from various Organization for Economic Cooperation and Development (OECD) countries. A chief limiting factor was the variety of languages spoken across these countries. We identified three languages that were spoken in many of the OECD countries—English, French, and German. In addition, we included Italian, Portuguese, and Spanish language papers because our research assistant happened to speak them. Using these languages, we were able to include 15 countries in our sample—Australia, Austria, Canada, France, Germany, Ireland, Israel, Italy, Japan, Luxembourg, New Zealand, Portugal, Spain, the United Kingdom, and the United States. For Japan and Israel, we used English-language versions of newspapers. For each country, we collected two newspapers, each relatively to the right or left of the other. We focused on papers that were mainstream rather than ideologically extreme or tabloid and that had a long time series of articles available.

The list of papers and coverage dates are given in table A.1 in the appendix, with the coverage ranging from 1977 to 2014.

Our data include 3 million articles from 30 newspapers retrieved from LexisNexis and ProQuest. From these articles, we constructed monthly measures of economic sentiment, growth sentiment, unemployment sentiment, and inflation sentiment, for each of the 30 newspapers. These measures are meant to reflect the tone that each newspaper uses to cover the economy in general, growth, the labor market, and inflation. We used dictionary-based coding to construct these measures, with initial dictionaries from WordStat (Péladeau 1998, 22–29) that we extended.

Consider a simplified example. A set of words is used to indicate positive sentiment, and another set of words is used to indicate negative sentiment. Another set of words is used to indicate economic growth. Positive growth sentiment is calculated as the proportion of positive words near growth words, as a fraction of the proportion of positive and negative words near growth words. We used a slightly more sophisticated measure than this that took into account words that indicate increases (which were coded as positive) and decreases (which were coded as negative). We also accounted for negations (which would alter the meaning of nearby positive, negative, increasing, and decreasing words) and for words that indicated a recession (which we coded as negative growth sentiment). Similar rules were used to construct unemployment sentiment and inflation sentiment. Economic sentiment was coded based on the proximity of positive and negative sentiment to words indicating either growth, unemployment or inflation.

A small literature has arisen about the relative merits of dictionary and supervised learning methods with a consensus that both can be effective under given circumstances (Barberá et al. 2021; Dun, Soroka, and Wlezién 2021). Our application required a large sample of newspapers in multiple languages. Dictionary-based methods were easier to extend to six languages, mostly due to the difficulty of collecting a separate training set in each language. In previous work (Kayser and Peress 2021), we validated our measures, including comparisons to human-coded data in three of the six languages, finding that the amount of measurement error was relatively small for most newspapers. In addition, we demonstrated that the sentiment measure correlated with measures of the actual economy in expected ways. We report the full time series for growth sentiment in figure 1.

Economic data

Our two sources of economic data were the OECD and the International Monetary Fund (IMF). We used the highest frequency data that were available. If monthly data were

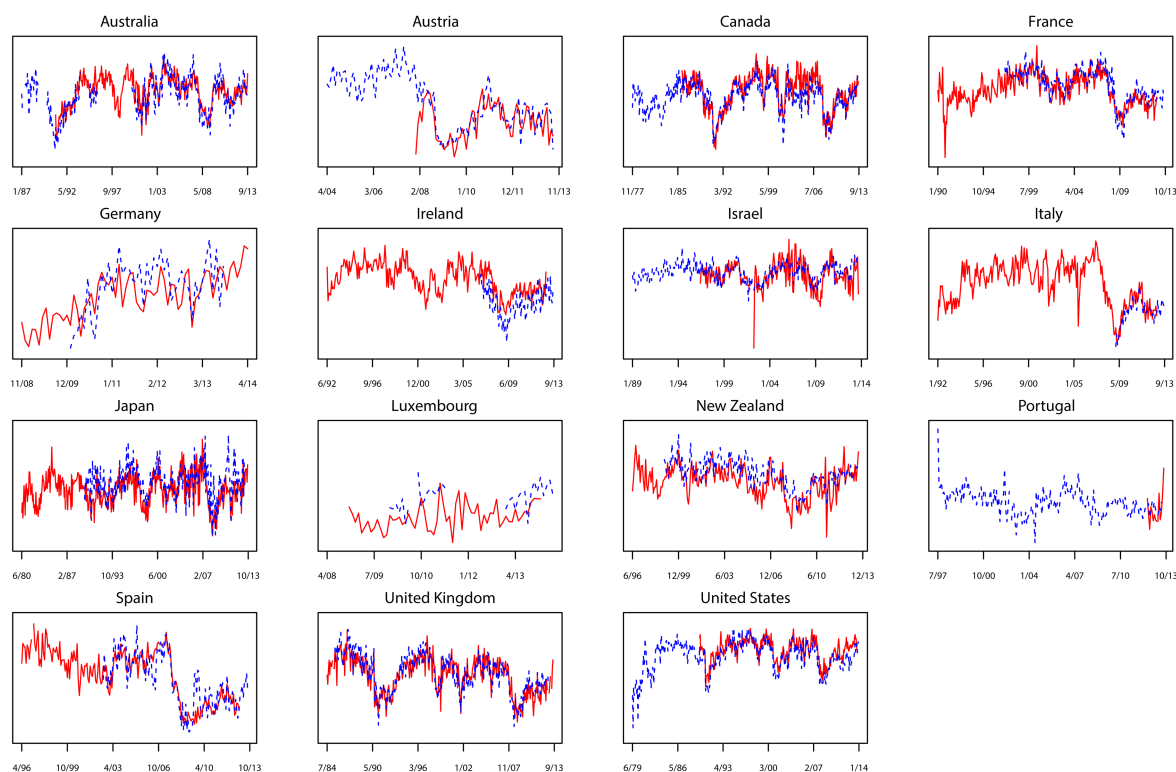


Figure 1. Comparing left-wing and right-wing sentiment. The figure reports the proportion of positive reports on the economy in left-wing (solid line) and right-wing (dashed line) newspapers over time.

available (as was often the case for unemployment and inflation), we used monthly data. If only quarterly data were available, we converted the quarterly data to monthly data as follows. For growth, unemployment, and inflation, we assumed a constant rate throughout the time period. When quarterly data were not available, we imputed the quarterly data based on yearly data, and we then imputed the monthly data based on the quarterly data. We used the highest level available preferentially, we used harmonized data (for unemployment and inflation) preferentially over unharmonized data, and we used the OECD data preferentially over the IMF data.

Once the data were converted to monthly values, we could then aggregate them to various other time periods. For example, consider an election held in March of 2003. In the election, voters may respond to unemployment in the current month, the current quarter, the current year, etc. Quarterly unemployment would be computed as average unemployment in the last three months. Yearly unemployment would be computed as the average unemployment in the last 12 months. However, suppose that only yearly unemployment was available. Although it may seem redundant to impute the monthly data based on yearly data only to convert the monthly data back to yearly data, the yearly data for March would be based on both the 2012 and 2013 unemployment rates, with the 2012

rate receiving a weight of 9/12 and the 2013 data receiving a weight of 3/12, so that the converting we do allows us to interpolate appropriate economic aggregates for elections that did not coincide with the reporting of economic aggregates.

Election results and vote intention

We assembled a dataset on democratic election results for the 15 OECD countries in our sample from a variety of sources. Our main sources were “The International Almanac of Electoral History” by Thomas T. Mackie and Richard Rose; “Elections in Europe: A Data Handbook” by Dieter Nohlen and Phillip Staver; “Elections in the Americas: A Data Handbook: Volume 1: North America, Central America, and the Caribbean” by Dieter Nohlen; “Elections in Asia and the Pacific: A Data Handbook: Volume 1: Middle East, Central Asia, and South Asia” by Dieter Nohlen, Florian Grotz, and Christof Hartmann; “Elections in Asia and the Pacific: A Data Handbook: Volume 2: South East Asia, East Asia, and the Pacific” by Dieter Nohlen, Florian Grotz, and Christof Hartmann; ParlGov (<http://www.parlgov.org/>); the Constituency-Level Elections Archive (<http://www.electiondataarchive.org/>); Adam Carr’s Election Archive (<http://psephos.adam-carr.net/>); and various countries’ Wikipedia pages. We used the election results dataset

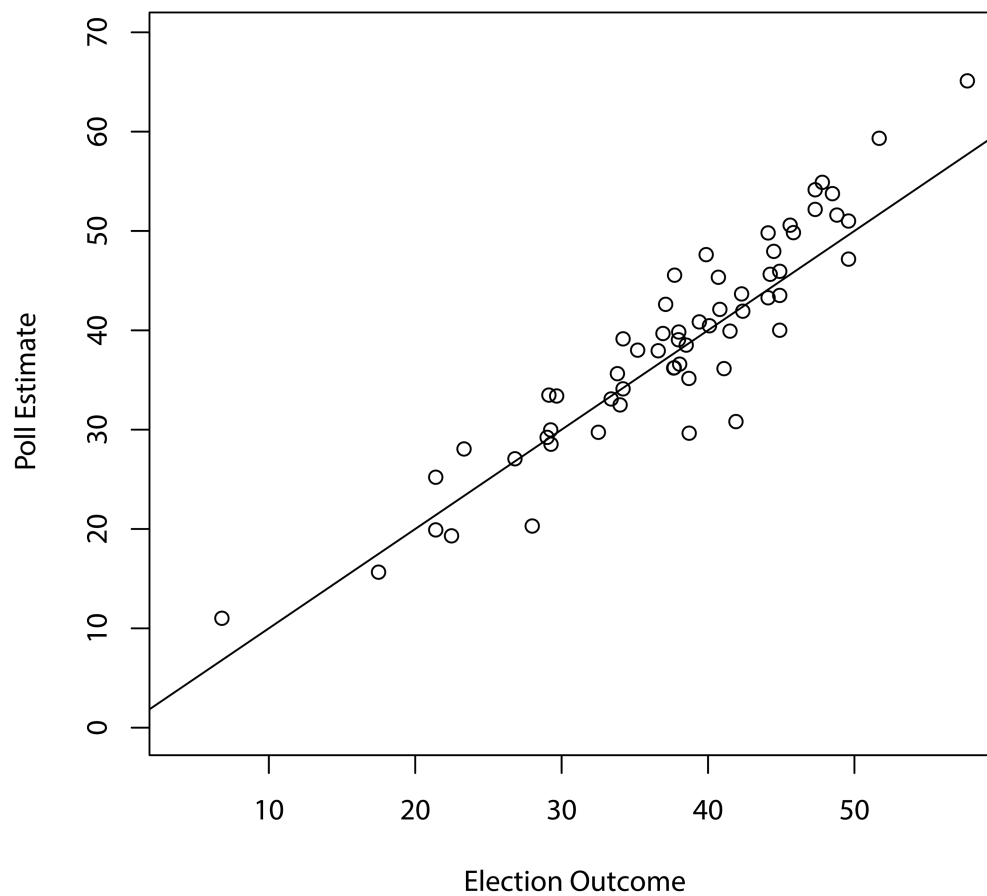


Figure 2. Incumbent vote share versus incumbent vote intent. The figure compares poll estimates to actual election outcomes to measure the accuracy of the vote intent measure.

to create one of our main dependent variables—the vote share of the incumbent prime minister’s party.

Because of the limited number of elections that intersect with the coverage of our media time series, we also collected survey estimates of voter intent. We used survey data from the Canadian Opinion Research Archive (for Canada), Elections Ireland (Ireland), the Eurobarometer (Austria, France, Germany, Ireland, Italy, Luxembourg, Portugal, Spain, and the United Kingdom), Gallup (the United States), ICM Research (the United Kingdom), Newspoll (Australia), Roy Morgan (Australia and New Zealand), and polls compiled by Wikipedia (Canada, Germany, Israel, Italy, Luxembourg, Portugal, and Spain). It was not possible to collect reliable vote intention data from Japan.⁵ The vote intent data is observed about 10 times more frequently than elections occur. The average country has an election a little less than once every four years,

whereas we observe a poll about every five months. Preelection polling for legislative elections has become more common over time, so in the earlier part of the sample, polling was observed less frequently, and in more recent years, polling is observed more frequently. In figure 2, we compare actual incumbent vote share to the poll-based estimate, in months where we observed both a poll and an election outcome. The results indicate a close correspondence. The R^2 is 0.86, and the standard error of regression is 3.97. The latter is close to what we would expect due to sampling error.

MEDIATION ANALYSIS

Do voters vote based on the economy or based on media reports of the economy? To the extent we observe a correlation between media reports of the economy and the vote share of the incumbent prime minister’s party, is this because voters are affected by the newspaper coverage or because the newspaper coverage simply provides a proxy for the actual state of the economy? To investigate this, we perform a mediation analysis (Acharya, Blackwell, and Sen 2016; Baron and Kenny 1986).

5. In Japan, public opinion polls are inaccurate because many respondents refuse to respond when they do not support the incumbent government.

For the mediation analysis, we estimated the following structural equations:

$$V_n = \beta_0 + \beta_1 G_n + \beta_2 U_n + \beta_3 I_n + \beta_4 GS_n + \beta_6 US_n + \beta_7 IS_n + \epsilon_n, \quad (4)$$

$$GS_n = \gamma_0^G + \gamma_1^G G_n + \eta_n^G, \quad (5)$$

$$US_n = \gamma_0^U + \gamma_1^U U_n + \eta_n^U, \quad (6)$$

$$IS_n = \gamma_0^I + \gamma_1^I I_n + \eta_n^I, \quad (7)$$

where V_n indicates incumbent vote share, G_n indicates growth, U_n indicates unemployment, I_n indicates inflation, GS_n indicates growth sentiment, US_n indicates unemployment sentiment, and IS_n indicates inflation sentiment. We can plug equations 5, 6, and 7 into equation 4 to obtain

$$V_n = \beta_0 + (\beta_1 + \beta_4 \gamma_1^G) G_n + (\beta_2 + \beta_5 \gamma_1^U) U_n + (\beta_3 + \beta_6 \gamma_1^I) I_n + \epsilon_n + \beta_4 \eta_n^G + \beta_5 \eta_n^U + \beta_6 \eta_n^I. \quad (8)$$

Here, we have that $\beta_1 + \beta_4 \gamma_1^G$, $\beta_2 + \beta_5 \gamma_1^U$, and $\beta_3 + \beta_6 \gamma_1^I$ represent the total effects of growth, unemployment, and inflation; β_1 , β_2 , and β_3 represent the direct effects; $\beta_4 \gamma_1^G$, $\beta_5 \gamma_1^U$, and $\beta_6 \gamma_1^I$ represent the indirect effects; and $\epsilon_n + \beta_4 \eta_n^G + \beta_5 \eta_n^U + \beta_6 \eta_n^I$ is an error term.

We report results for the regressions described in table 1.⁶ In column 1, we consider vote share as the dependent variable. We see that growth and unemployment are statistically significant and have the expected signs, whereas inflation is not statistically significant.⁷ The sample here is restricted to the 15 OECD countries in our newspaper sample. If we were to further restrict this to elections for which we observe newspaper data, the sample would shrink even more. To overcome this problem, we collected survey data on vote intention, which, as we showed in figure 2, was highly correlated with the vote and was available at greater frequency, increasing our sample size. Column 2 presents the same specification, but with vote intention as the dependent variable. The effect of growth appears to increase somewhat, and the effect of unemployment appears to decrease, but the effects of growth and unemployment retain the expected signs and are statistically

significant. The analysis of these models demonstrates that economic vote effects emerge in our baseline specification.⁸ As mediation analysis decomposes this effect, it is important to demonstrate that we are decomposing an effect that actually exists.

Continuing toward the mediation analysis, column 3 employs vote share as the dependent variable and adds three variables derived from our automated text analysis—growth sentiment, unemployment sentiment, and inflation sentiment—to the specification. In each case, the sentiment variables are calculated as the average of the sentiment in the left-wing and right-wing paper in each country in each time period. The coefficients on sentiment are statistically insignificant.

We expand on the analyses in the final three models of table 1 in table 2 by calculating the direct, indirect, and total effects, along with their associated standard errors. These quantities can be computed in a number of ways, based on the coefficients of the structural model or sequential g-estimation. In the context of our linear model, the structural model and sequential g-estimation approaches yield equivalent estimates, however.⁹ Column 1 in table 2 presents the relevant mediation results for column 3 in table 1. We find that unemployment has a negative total and direct effect, while all other effects fail to achieve statistical significance. The ratio of the indirect effect to the total effect for unemployment is precisely estimated, very close to zero, and statistically indistinguishable from zero, suggesting that the effect of unemployment on the vote is not mediated through news sentiment. The effect of growth is indeterminant.

To address this indeterminacy, we repeated the analysis using vote intention in column 4 of table 1 and column 2 of table 2. Here, we find that growth has both a direct and indirect effect on vote intention, unemployment has a direct effect, and inflation appears to have no effect.¹⁰ Moreover, the point estimates suggest that the direct effect of growth is larger than the indirect effect. There is, however, a reason to believe that our analysis may depress the size of the indirect effect of growth. Voters presumably respond to media coverage broadly, but we only have measures of sentiment from two

6. We report robustness checks in app. A.2

7. In their survey of the literature, Lewis-Beck and Stegmaier (2013) argue that inflation has a small effect on the vote except during periods when it is particularly salient. Consistent with this, when we expanded our data set to 24 OECD countries, we found a small but statistically significant effect for inflation.

8. Hellwig and Samuels (2008), for example, estimate a similar coefficient of 0.55 on GDP growth for a large sample of high clarity countries.

9. Acharya et al. (2016) develop approaches that allow for certain additional interactions in the case of a single treatment and single mediator, which can be used if these interactions are theoretically indicated. In app. table A.2, we compute the Average Controlled Direct Effect (ACDE) using sequential g-estimation. The ACDE is numerically identical to the direct effect we report, and the standard errors are nearly identical, with the differences due to the fact that the standard error we compute for the ACDE relies on the bootstrap.

10. The indirect effect of inflation is marginally significant.

Table 1. Elections Outcomes, the Economy, and Sentiment

	(1)	(2)	(3)	(4)	(5)
Dependent Variable:	Vote Share	Vote Intent	Vote Share	Vote Intent	Vote Intent
Independent variables					
Constant	42.255*** (1.723)	38.215*** (.483)	60.842*** (10.484)	32.251*** (1.348)	27.259*** (3.423)
Growth (yearly)	.647** (.204)	1.065*** (.079)	.823 (.517)	.561*** (.120)	.552*** (.117)
Unemployment (yearly)	−.868*** (.196)	−.116** (.040)	−1.047*** (.314)	−.162** (.054)	−.195** (.065)
Inflation (yearly)	−.021 (.136)	−.077 (.066)	.078 (.415)	.080 (.124)	−.159 (.123)
Growth sentiment			−13.397 (10.743)	8.355*** (1.674)	7.539* (3.161)
Unemployment sentiment			−5.003 (8.106)	1.506 (2.056)	6.566 (6.515)
Inflation sentiment			−20.271+ (12.173)	3.498* (1.715)	11.575** (4.351)
Estimator	OLS	OLS	OLS	OLS	IV
Number of months	202	2,445	95	1,708	2,452
Number of countries	15	15	15	15	15
R ²	.203	.095	.191	.078	

Note. Heteroskedasticity robust standard errors in parentheses. Column 5 corrects the results of column 4 for measurement error using instrumental variables. IV, instrumental variable; OLS, ordinary least squares.

+ $p < .10$.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

media outlets in each country. We show that this problem can be categorized as measurement error, and we correct for this problem.

Measurement error

Theoretically, we are interested in the effect of sentiment in the overall media environment on the economic vote. Our measure departs from this because we measure the average sentiment in a sample of two media outlets, which introduces measurement error. The amount of measurement error will be determined by how strongly sentiment is correlated across media outlets. In one extreme—if sentiment is perfectly correlated across media outlets—using a single outlet as a measure of overall sentiment will not result in measurement error. On the other hand, if sentiment in media outlets is weakly correlated, there will be substantial measurement error that will lead to attenuated estimates of the effect of media sentiment.

It is possible to quantify the degree of measurement error since we observe multiple measures of sentiment in each country and time period. Let s_{cjt} denote the sentiment for paper j in country c in time period t , and let μ_{ct} denote the true sentiment in the media environment. Assume that for each paper, $s_{cjt} = \mu_{ct} + \epsilon_{cjt}$, where ϵ_{cjt} are independent errors. Our measure of overall sentiment is $\bar{s}_{cj} = \frac{1}{2}(s_{c1t} + s_{c2t})$. We can measure the reliability of $\bar{s}_{cj}$ as a measure of μ_{ct} by calculating the ratio of variances. It can be demonstrated that¹¹

$$R = \frac{2\text{Cov}(\hat{s}_{c1t}, \hat{s}_{c2t})}{\text{Var}(\hat{s}_{c1t}) + \text{Cov}(\hat{s}_{c1t}, \hat{s}_{c2t})}. \quad (9)$$

Using estimates for the variance of sentiment, $\text{Var}(\hat{s}_{cjt})$, and the covariance between media outcomes, $\text{Cov}(\hat{s}_{c1t}, \hat{s}_{c2t})$, we can

11. Note that, $\text{Var}(\bar{s}_{cj}) = \frac{1}{4}(2\text{Var}(\epsilon_{cjt}) + 4\text{Var}(\mu_{ct}))$, $\text{Cov}(\hat{s}_{c1t}, \hat{s}_{c2t}) = \text{Cov}(\mu_{ct} + \epsilon_{c1t}, \mu_{ct} + \epsilon_{c2t}) = \text{Var}(\mu_{ct})$, and $\text{Var}(\hat{s}_{cjt}) = \text{Var}(\mu_{ct}) + \text{Var}(\epsilon_{cjt})$. This means that $\text{Var}(\epsilon_{cjt}) = \text{Var}(\hat{s}_{cjt}) - \text{Cov}(\hat{s}_{c1t}, \hat{s}_{c2t})$ and that $\text{Var}(\bar{s}_{cj}) = \frac{1}{2}(\text{Var}(\hat{s}_{c1t}) + \text{Cov}(\hat{s}_{c1t}, \hat{s}_{c2t}))$, which leads to the formula for the reliability.

Table 2. Mediation Analysis

	(1)	(2)	(3)
Dependent Variable:	Vote Share	Vote Intent	Vote Intent
Growth			
Total effect	.470 (.389)	.781*** (.108)	.751*** (.137)
Direct effect	.823 (.517)	.561*** (.120)	.552*** (.117)
Indirect effect	-.353 (.283)	.220*** (.044)	.199* (.083)
Indirect/total ratio	-.751 (.794)	.282 (.071)	.265 (.095)
Unemployment			
Total effect	-1.060*** (.312)	-.159** (.053)	-.179** (.057)
Direct effect	-1.047*** (.314)	-.162** (.054)	-.195** (.065)
Indirect effect	-.012 (.020)	.004 (.005)	.016 (.016)
Indirect/total ratio	.012 (.020)	-.024 (.033)	-.091 (.085)
Inflation			
Total effect	.131 (.410)	.071 (.124)	-.189 (.122)
Direct effect	.078 (.415)	.080 (.124)	-.159 (.123)
Indirect effect	.053 (.033)	-.009 ⁺ (.005)	-.030* (.012)
Indirect/total ratio	.406 (1.322)	-.130 (.237)	.161 (.126)
Estimator	OLS	OLS	IV

Note. Standard errors calculated using the delta method in parentheses. Results in columns 1, 2, and 3 are calculated based on columns 3, 4, and 5 of table 1, respectively. The first column reports mediation analysis for vote share, the second reports mediation analysis for vote intent, and the third corrects the results in column 2 for measurement error using instrumental variables. In each case, the direct effect measures the effect of the economic variable controlling for sentiment and the indirect effect measures the effect of the economic variable as mediated by sentiment. IV, instrumental variable; OLS, ordinary least squares.

⁺ $p < .10$.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

calculate that the reliability is 83.3% for economic sentiment, 87.2% for growth sentiment, 52.7% for unemployment sentiment, and 52.3% for inflation sentiment.

As can be seen, the estimate for growth sentiment is quite reliable. Unemployment and inflation sentiment are less reliable. It is possible to correct our estimates for measurement error, which we do in column 5 of table 1 and column 3 of table 2. The correction for measurement error is based on

instrumental variables (Greene 2000).¹² The results, correcting for measurement error, continue to suggest that growth has both a direct and indirect effect, and the indirect effect is

12. In the instrumental variable-based correction, we instrument for growth sentiment from one newspaper using the growth sentiment from the other newspaper in that country. We similarly instrument for unemployment and inflation sentiment.

approximately the same size. Unemployment continues to have only a direct effect. We observe a statistically significant indirect effect for inflation. The evidence for an effect of inflation is weaker since the total effect is not statistically significant.

SUBSTANTIVE EFFECTS

To get a sense of the effect sizes, we used the model from column 5 of table 1 to gauge the effect of a 1-standard-deviation improvement of the various economic statistics and media sentiment on the intent to vote for the incumbent party. A 1-standard-deviation improvement in growth, unemployment, and inflation, holding media sentiment constant, would lead to an increase in the incumbent's vote share of 1.7, 0.8, and 1.0 percentage points, respectively. Improving all three simultaneously would lead to an increase of 3.5 percentage points. Improving the economic variables would also have an effect that would work through media sentiment—these effect sizes would be 0.6, 0.1, and -0.2 , respectively, or 0.5 if all three were simultaneously improved. We could instead consider the effect of improving media sentiment itself by 1 standard deviation. Improving growth sentiment, unemployment sentiment, and inflation sentiment by 1 standard deviation would lead to an increase in vote intention of 1.0, 0.7, and 1.2 percentage points, respectively, or 2.9 percentage points if all three were simultaneously improved. Improving all three measures of the economy and allowing these effects to carry through sentiment would lead to a 4.0 percentage point increase in vote intention. Improving all three measures of the economy and all three measures of sentiment by 1 standard deviation each would lead to a 6.3 percentage point increase in vote intention.

The analysis leaves open the possibility of moderately sized media effects—improving all three measures of media sentiment by 1 standard deviation would increase the incumbent's vote share by only about 3 percentage points. However, media sentiment and the economy are unlikely to be misaligned to this extent. To illustrate this, we compared the predicted vote share of the incumbent party when growth sentiment, unemployment sentiment, and inflation sentiment were the same number of standard deviations from their mean as growth, unemployment, and inflation. We compared this to the predicted vote share of the incumbent party when all six variables were held at their observed values. This change can thus be thought of as representing the gain in vote share that the incumbent party would receive from accurate coverage. In the average case, the vote share of the incumbent decreased by 0.15 percentage points with 80% of the observations falling between a loss of 1.84 percentage points and a gain of 1.49 percentage points.

Although our data show that economic sentiment in most newspapers tracks that of their ideological counterparts and the economy fairly faithfully and that the mediating effect of economic reporting on the economic vote is limited, deviations of media reporting from actual economic performance are corrected with election outcomes. To show the utility of our exercise for the understanding of individual campaigns and elections, we review one case in which the difference between newspaper reporting and the actual economy was large enough to determine a narrow win and one case in which, despite assertions to the contrary, it most likely was not.

Saved by the media? The 2010 Australian federal election

In 2010, the Australian Labor Party competed for a second consecutive term in office under the leadership of Julia Gillard. Labor, during the previous government of Kevin Rudd, had presided over the economy during the global financial crisis that proved to be shorter and less severe in Australia than in many other developed countries.

Figure 3 plots out standardized rate of gross domestic product (GDP) growth (solid line) and standardized growth sentiment in two newspapers, *The Age* (dashed line) and *The Herald Sun* (dotted line). After a steep contraction in 2008, Australian growth levels returned to near, albeit slightly below, their long-run average. The news media, however, which had been slow to respond to the economic recovery in their economic reporting, provided overly positive coverage for the full year preceding the election—the standardized media sentiment exceeded the standardized rate of growth by an average of 0.7 standard deviations in the half-year preceding the election. From the media effect estimates in table 3, we estimated that this resulted in a 0.6 percentage point increase in the incumbent Labor Party vote share over what would have been expected if reporting on economic growth matched actual economic growth. The election proved to be very close, with Gillard's Labor Party, after instant run-off vote allocations, defeating Tony Abbott and the Liberal/National coalition by a margin of 50.12 to 49.88—considerably less than the 0.6 points gained from beneficial media coverage of economic growth. When an election is so close, many things can tip the outcome. In this case, overly positive economic reporting may count as one of them.

False positive? George H. W. Bush's reelection campaign

Former President George H. W. Bush famously attributed his reelection loss to inaccurate media coverage of the economy. Here we consider the correctness of Bush's assessment and

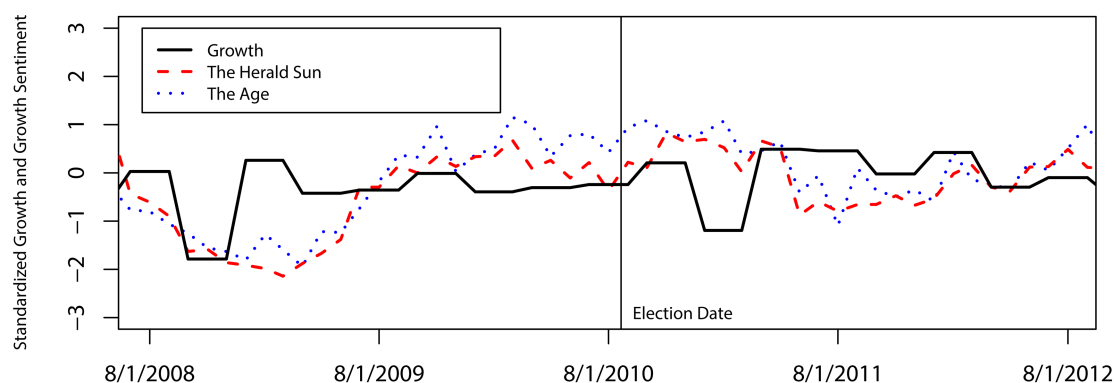


Figure 3. Standardized monthly growth and standardized news sentiment prior to and following the 2010 Australian federal election.

whether the size of the effect of the media is large enough to explain his election loss. Figure 4 considers the four years surrounding Bush's second 1992 election. During the last year of President Bush's term, growth was slightly better than average. Sentiment in both the *New York Times* and the *Wall Street Journal*, however, was below average, converged to neutral during the election month, and became more positive than the actual economy after Bush lost reelection. Initially, this may seem roughly consistent with Bush's claim.

Our findings, however, yield a different conclusion—that the role of media coverage of the economy in Bush's loss was exaggerated. The actual economy makes up a large part of the economic vote. Media coverage of growth also has an effect but

a smaller one than growth itself on incumbent vote share—a 1-standard-deviation increase in growth sentiment, holding growth and the other covariates constant, would lead to an increase in incumbent vote share of about 1.0 percentage points. Late in his reelection campaign, standardized growth sentiment had converged to only half a point below standardized growth. Had standardized growth sentiment matched standardized growth, this would have netted Bush less than half of a percentage point—far too small to explain a more than 5 percentage point loss. The level of inaccuracy of economic coverage leading up the election is among the largest inaccuracies we observed in our sample: due to the limited role of the media in shaping the economic vote, attributing the entirety of Bush's loss to media coverage is an exaggeration.

Table 3. Effect Sizes

Economic Variable:	Growth	Unemployment	Inflation	All
Direct effect	1.696*** (.358)	.783** (.261)	.995 (.771)	3.473*** (.848)
Indirect effect	.610* (.256)	.065 (.066)	-.190* (.077)	.485+ (.255)
Media effect	.983* (.412)	.701 (.696)	1.179** (.443)	2.863*** (.720)
Direct + indirect	2.306*** (.419)	.848** (.304)	.804 (.789)	3.958*** (.855)
Direct + media	2.679*** (.519)	1.484+ (.877)	2.174** (.813)	6.336*** (1.030)

Note. Effect sizes on intention to vote for the incumbent's party are measured for a 1-standard-deviation improvement in growth, unemployment, inflation, and all three economic statistics simultaneously. Estimates are calculated based on column 5 of table 1 and columns 1, 2, and 4 of app. table A.3. Standard errors calculated using the delta method in parentheses.

+ $p < .10$.

* $p < .05$.

** $p < .01$.

*** $p < .001$.

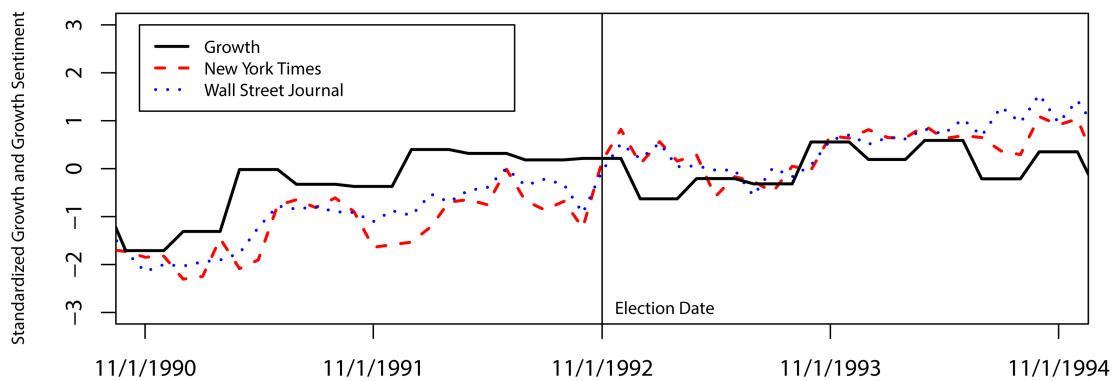


Figure 4. Growth, growth sentiment, and Bush's reelection campaign. Standardized monthly growth (solid line), *New York Times* growth sentiment (dashed line), and *Wall Street Journal* growth sentiment (dotted line).

CONCLUSION

What are the mechanisms behind the economic vote? Are voters swayed by objective economic conditions? Or are they swayed by how the media colors their reporting on economic conditions? The nature of how voters acquire information about the economy matters because it speaks to the degree to which the media can influence election outcomes and to whether voters can hold elected officials responsible for economic conditions. Having conducted the first mediation analysis of the economic vote, we are uniquely able to estimate how large a role deviations in media sentiment play.

This article's main finding is that deviations in economic news sentiment from the objective performance of the economy play only a modest role in the economic vote. We find that the effect of economic growth, but not unemployment, is mediated by the tone of newspaper reporting. About 30% of the effect of growth on incumbent party vote/vote intention is mediated by the tone of news coverage. Inflation sentiment, for its part, does not appear to exert a strong mediating effect in our sample, with a small indirect effect emerging in some but not in other specifications.

Our analysis leaves room for media slant to effect election outcomes but places limits on it. The economy itself has a moderate impact on election outcomes—we estimated that simultaneously improving growth, unemployment, and inflation by 1 standard deviation would increase the vote share of the incumbent party by about 4 percentage points. Simultaneously improving growth, unemployment, and inflation sentiment by 1 standard deviation would increase its vote share by about 3 percentage points. Empirically, however, because media sentiment tracks the objective economy very closely, it rarely changes outcomes by more than a percentage point or two. Inaccurate media coverage is therefore unlikely to change anything but a very close election. To be sure,

something similar could be said about the economic vote itself, but the effects of the economy are somewhat larger and media coverage of the economy is largely accurate.

One could argue that 30% is a conservative estimate, given our analysis's focus on one type of media (newspapers) and one part of the market (mainstream newspapers). For other news sources to add to this effect, however, their economic news sentiment would need to be at least partly uncorrelated with that of mainstream newspapers and reach a nontrivial proportion of voters. This is plausible but there are also reasons to doubt that additive effects from other news outlets are large. Research has found that media tone in economic news tracks the economy (Boydston et al. 2018) and sentiment in other newspapers (Kayser and Peress 2021) with considerable fidelity, that newspapers set the agenda for other news media (Roberts and McCombs 1994), and that the content in newspapers differs little from that in other sources (Druckman 2005; Maier 2010).

Our results also relate indirectly to the debate over socio-tropic versus pocketbook voting (Kiewiet 1983). According to pocketbook voting, voters punish incumbent governments based on their personal financial situation, whereas socio-tropic voting holds that voters punish incumbents based on their perceptions of the broader economy. The information that voters glean from media coverage informs them about the economy more broadly, so to the extent that the economic vote is mediated by news coverage, it is not pocketbook voting. The reverse correspondence does not hold, however. Personal experience can comprise losing one's job or getting a raise—which represents a voter's personal financial situation—but it can also involve hearing about a friend losing a job or seeing a business close, which may inform the voter's sense of the broad economy. Had we found evidence for an economic vote that was primarily mediated by news coverage, this would

present a challenge for pocketbook voting. Our finding of strong direct effects for both growth and unemployment is consistent with both sociotropic voting and pocketbook voting.¹³

Deviations in economic news sentiment from the actual performance of the economy can change election outcomes, but rarely by more than a percentage point or two, as we illustrated using the cases of the 1992 US presidential and 2010 Australian general elections. This is good news for electoral accountability. When voters punish or reward elected officials in accordance with news sentiment that does not match the performance of the actual economy, they weaken the control of the governed over those who govern and introduce the possibility of media manipulation of the economic vote. We demonstrated that although deviations of economic news sentiment from the objective economy are correlated with incumbent vote shares, the substantive size of these effects is typically small. Politicians and partisans often blame the media for poor results in polls and elections. At least in the case of economic news, inaccurate or biased media sentiment is seldom truly to blame.

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13. App. A.4 provides an expanded discussion of how our results relate to the sociotropic vs. pocketbook voting debate.

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